Plant Protection

Land Sciences

May, 2026

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Plant Protection (A08221)

Short Title: Plant Protection
Faculty Science and Computing
Department Land Sciences
Cluster Horticulture
Author CBYRNE
Credits 5 **Level:** Introductory

Description of Module / Aims

The module will enable the student to describe and apply techniques to control weed, pests and diseases. The student will learn how to calibrate and use a knapsack sprayer correctly.

Programmes

HORT-0002	BSc in Horticulture (WD_SHORT_D)	1 / 1 / M
HORT-0002	BSc in Horticulture (WD_SHORT_DX)	1 / 1 / M

Indicative Content

- Classification of pests, diseases, disorders and weeds
- Identification of pests, diseases, disorders and weeds
- Effect of pests, diseases, disorders and weeds on plant growth
- Control measures and IPM techniques for minimising infestations of pests, diseases, disorders and weeds, sustainably in crops
- Main types, application methods, formulations and modes of action of pesticides, plant pesticide resistance
- Outline the risks and benefits associated with pesticide use including plant pesticide resistance, methods of reducing contamination of the environment, and routes of contamination of the body, and the appropriate use of PPE when working with pesticides
- Current EU and National Legislation including Statutory Instrument `Good Plant Protection Practice' (GPPP) in relation to pesticides. Maintenance of records, and safe handling, storage, transport, use and disposal of unused pesticides, empty containers, and application equipment (i.e. a `system of work').
- Carry out risk assessment, and development of strategies for dealing with accidents and incidents involving spillages, contamination and poisoning
- Calibration, mixing and correct use and storage of a knapsack sprayer, slug pellet operator, and weedlick. Interpretation of pesticide labels and Safety Data Sheets (SDS).

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify a range of horticultural pests, diseases, disorders and weeds.
2. Describe the effects of infestations of pests, diseases, disorders and weeds on crops.
3. Describe life cycles, monitoring and control methods for pests, diseases, disorders and weeds.
4. Describe the main types, formulations, modes of action, and resistance to pesticides.
5. Describe the current pesticide regulations.
6. Operate a knapsack sprayer, calculate and make up spray mix, correctly apply a pesticide, demonstrate effective cleaning, storage, and calibration of equipment.

Learning and Teaching Methods

- Practical demonstrations
- Lectures

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	2,3,4,5
Continuous Assessment	50%	
Practical	15%	1
Practical	35%	6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out practical tasks.

40%-49%: Demonstrates a limited knowledge of the subject matter. Shows some technical ability/skill in carrying out and reporting on practical tasks.

50%-59%: Demonstrates satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical tasks.

60%-69%: Demonstrates sound knowledge of the subject matter.

70%-100%: Demonstrates imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical tasks.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- O'Dwyer, J. _Plant Protection Workbook_. Ireland: Teagasc, 2013.

Supplementary Material

- "Resistance Management." http://www.agf.gov.bc.ca/pesticides/a_4.htm
- Buczacki, S and K Harris. _Pests, Disease and Disorders of garden plants_. London: Collins, 2000.
- Halstead, A. and P. Greenwood. _RHS Pests and Diseases_. London: Dorling Kindersley, 2009.
- Hessayon, D.G. _The Pest and Weed Expert_. London: Expert Books, 2009.
- Heyler, N., N. Cattlin and K. Brown. _Biological Control in Plant Protection_. 2nd ed. New York: CRC press, 2014.
- Stephens, R.J. _Theory and Practice of Weed Control_. Bath University, UK: Gage Distribution Co., 1982.

Requested Resources

- Lecture Room: Loose Seated